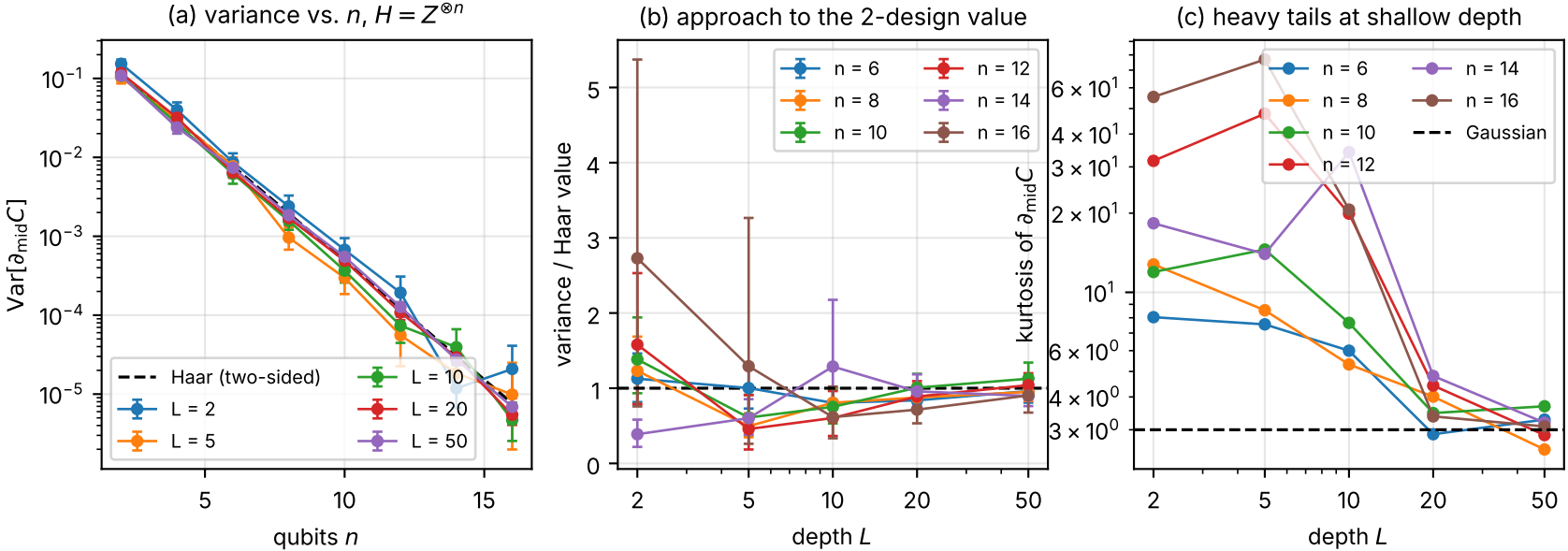
(a) variance vs. n, $H = Z^{\otimes n}$
$\mathrm{Var}[\partial_{\mathrm{mid}} C]$
qubits n
Haar (two-sided)
L = 2
L = 5
L = 10
L = 20
L = 50
(b) approach to the 2-design value
variance / Haar value
depth L
n = 6
n = 8
n = 10
n = 12
n = 14
n = 16
kurtosis of $\partial_{\mathrm{mid}} C$
(c) heavy tails at shallow depth
depth L
n = 6
n = 8
n = 10
n = 12
n = 14
n = 16
Gaussian